

RepMate — COMP90018 Assignment 2

Mobile Computing Systems Programming · Group [FILL: group #]

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Cover page

Subject: COMP90018 Mobile Computing Systems Programming **Assignment:** Assignment 2 — Group Project
Group number: [FILL: group #] **Application:** RepMate — a motion-sensing exercise form coach **Repository:** qichongchen/RepMate-COMP90018, submitted at commit [FILL: final SHA on main at submission]

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Publicity authorisation — decide before submitting. The brief invites the following statement. It is included here so the group can consciously keep or delete it; it is not yet agreed.

“We authorise the University of Melbourne to use material from our submission for publicity.”

1. Demo video

Link: [FILL: YouTube link]

The index below is built from the implemented feature set, so the video can be shot against it and every rubric criterion has a moment a marker can jump to. Timestamps are [FILL] until the video is cut.

#	Segment	Rubric criterion it evidences	Timestamp
1	App launch, sign in with Google, display-name claim	Connectivity	[FILL]
2	Onboarding and the exercise tutorial dialog	UI — Flow, UI — Language	[FILL]
3	Squat calibration: five prompted reps, profile derived	Sensors, Technical depth	[FILL]
4	Live squat workout: counter, haptic buzz, spoken count	Sensors, Responsiveness	[FILL]
5	Deliberate wobble that is not counted	Sensors (false-positive rejection)	[FILL]
6	Jumping jacks with the metronome	Sensors, Innovation — Novelty	[FILL]
7	Camera push-up set: pose overlay, stability gate	Technical depth, Innovation — Tech Knowledge	[FILL]
8	Post-workout summary: per-rep scores and reasons	Technical depth, UI — Language	[FILL]
9	Motion Replay: smoothed curve against calibration band	Innovation — Surprise	[FILL]
10	History, score-trend chart with depth line	UI — Appeal, UI — Reactiveness	[FILL]
11	Aeroplane mode: finish a workout, show it saved	Connectivity (offline-first)	[FILL]
12	Reconnect, show the leaderboard row appear	Connectivity (reconciliation)	[FILL]
13	Friend request sent, accepted on a second device	Connectivity (real-time listeners)	[FILL]
14	Ghost duel against a friend's best set	Innovation — Novelty, Impact	[FILL]
15	Safety check-in: notification, then escalation SMS	Innovation — Impact	[FILL]
16	Dark/light theme toggle, TalkBack on one screen	UI — Guidelines	[FILL]

[!GAP] Segment 16's TalkBack demonstration has not been rehearsed, and no TalkBack pass has been done on the app (see §5.7). Either rehearse it or drop the segment — do not film a screen reader session cold.

2. Build and run instructions

Written for a marker starting from a clean machine. Every version below is taken from the build files, not from memory.

2.1 Prerequisites

Requirement	Version	Where it is pinned
JDK	17	app/build.gradle.kts → compileOptions; CI uses Temurin 17
Android Gradle Plugin	9.3.2	gradle/libs.versions.toml
Gradle	9.5.0	gradle/wrapper/gradle-wrapper.properties (wrapper supplied — do not install Gradle)
Kotlin	2.2.10	gradle/libs.versions.toml
Android SDK	compileSdk 37 , targetSdk 37, minSdk 26 (Android 8.0)	app/build.gradle.kts
Android Studio	any release bundling AGP 9.3.x support	—
Node.js	20+	only needed for the Firestore rules tests

2.2 One required secret

app/google-services.json **is committed to the repository**, so no Firebase file needs to be supplied. One value is not committed:

in the repository root, create or edit Local.properties

```
echo "GOOGLE_WEB_CLIENT_ID=<OAuth Web client ID, client_type 3>" >> local.properties
```

The value is the client_id with "client_type": 3 inside app/google-services.json. The build **fails with an explicit message** if it is missing, rather than silently substituting a placeholder that would break Google Sign-In at runtime:

Missing GOOGLE_WEB_CLIENT_ID in local.properties.

Copy the OAuth Web Client ID (client_type 3) from app/google-services.json.

local.properties is git-ignored, which is why this step exists.

2.3 Build

```
git clone https://github.com/qichongchen/RepMate-COMP90018.git
```

```
cd RepMate-COMP90018
```

add GOOGLE_WEB_CLIENT_ID to Local.properties as above

```
./gradlew assembleDebug # Windows: gradlew.bat assembleDebug
```

The APK lands at app/build/outputs/apk/debug/app-debug.apk.

2.4 Run

On a physical device (recommended — this is a motion sensor app). Enable developer options and USB debugging, connect, then:

```
./gradlew installDebug
```

or, if Gradle cannot see the device (e.g. wireless debugging):

```
adb install -r app/build/outputs/apk/debug/app-debug.apk
```

On an emulator. Everything except the accelerometer exercises works. Squats and jumping jacks depend on

real device motion that the emulator's virtual sensors cannot reproduce usefully; push-ups need a camera, which the emulator can map to a webcam. Use a Pixel system image at API 34 or above.

Permissions requested on first use — each is requested at the point of use, not at launch:

Permission	Requested when	Consequence of refusing
CAMERA	starting a push-up workout	push-ups unavailable; squats and jumping jacks unaffected
POST_NOTIFICATIONS	enabling safety check-in	check-in reminders do not appear
SEND_SMS	enabling safety check-in	escalation SMS cannot be sent
ACCESS_FINE_LOCATION	enabling safety check-in	escalation SMS omits the map link

No permission is required to count squats or jumping jacks: the accelerometer and gyroscope are not protected permissions on Android.

2.5 Run the tests

```
./gradlew testDebugUnitTest           # 426 JVM unit tests, no device needed
./gradlew compileDebugAndroidTestKotlin # compiles the instrumented suite
./gradlew connectedDebugAndroidTest    # 47 instrumented tests; needs a device AND the emulators
npx firebase emulators:exec --only firestore "node --test tests/firestore.rules.test.cjs"
```

The instrumented suite additionally needs the Firebase emulators reachable from the device and disposable test credentials in `local.properties` (`TEST_ACCOUNT_A_EMAIL`, `..._PASSWORD`, `TEST_ACCOUNT_B_EMAIL`, `..._PASSWORD`, `TEST_ACCOUNT_B_UID`). On a physical device the emulator ports must be reversed: `adb reverse tcp:8080 tcp:8080 && adb reverse tcp:9099 tcp:9099`. Markers short of time should run the first and last commands, which need no device at all.

3. Compilation screenshot

[INSERT: Android Studio console screenshot showing successful build]

Produce it with a clean build, so the console shows the whole task graph rather than a wall of UP-TO-DATE:

```
./gradlew clean  
./gradlew assembleDebug testDebugUnitTest --console=plain
```

Screenshot the Android Studio **Build** tool window showing BUILD SUCCESSFUL, with the test task included so the 426 passing unit tests are visible in the same frame.

4. Itemised contributions

272 commits on main, 2026-08-25 to 2026-10-10. Each row names what that member **created and then maintained**, recoverable with `git log main --diff-filter=A` under `.mailmap`.

Member	What they built	Com- mits
Mohit Nanda	Motion engine and the data feeding it. SensorSource (50 Hz fusion, gravity retained, boot-clock timestamps), SquatRepDetector, JumpingJackRepDetector, BurstDetector, CalibrationProfile, Models. The trace corpus : all six recordings and their <code>.expect</code> ground truth, plus the TraceLibrary/TraceLoader auto-discovery that makes adding one cost nothing. The push-up camera path end to end – PushupPoseAnalyzer, PushupFrameProcessor, PushupWorkoutScreen/ViewModel, PushupAngleTrace, Throttle – with its gating (ArmLock, CountingGate, PoseGeometry, PhoneStabilityGate). The safety check-in package, 15 files. Leaderboard scoring and publication (LeaderboardPoints, LeaderboardSync, LeaderboardWriter). Largest share of the test suite: 59 commits under <code>app/src/test</code> .	80
Henrico Leodra	The presentation layer. Navigation (NavGraph, 22 commits; StartupViewModel), the theme, and the shared component set (RepMateCard, RepMateButton, BottomNavBar, ChartAxes, EmptyState, ExerciseChip, LoadingOverlay). Screens: Welcome, Log in, Sign up, Forgot password, Onboarding, Choose display name, Home, History, Session detail, Motion replay, Profile, Ghost Duel, Live Workout, Calibration, tutorial dialogs. The authentication flow (AuthViewModel, AuthValidation, PasswordResetCooldownTracker) and display-name claim (DisplayNameRules, FirestoreUsernameRepository). Guest history migration – GuestHistoryMigrator and its 30 tests, the project's largest test file – plus LocalBestScoreSync. Also RepFeedback (haptics, text-to-speech) and the metronome.	67
Qichong Chen (Jasper)	Project foundations and the social read path. Created the Gradle build (<code>build.gradle.kts</code> , <code>app/build.gradle.kts</code> , <code>libs.versions.toml</code>), MainActivity, RepMateApplication, the Room database class, DatabaseMigrations and DatabaseModule – the scaffolding every other workstream was built on. Owns the Friends UI (FriendsScreen, FriendsViewModel) and the leaderboard read path (LeaderboardScreen, LeaderboardViewModel, LeaderboardRepository, FirestoreLeaderboardRepository). Heaviest contributor to <code>res/</code> (19 commits) and build configuration (18). Wrote <code>SessionDaoTest</code> ; contributed to <code>firestore.rules</code> and its test suite.	46
Xiaonuo Jia (Lisa)	Persistence, cloud and security. Set up Firebase (<code>google-services.json</code> , <code>firebase.json</code> , <code>.firebaserc</code>) and wrote <code>firestore.rules</code> with <code>tests/firestore.rules.test.cjs</code> , the 68-test emulator suite covering it. The complete Room layer : <code>SessionDao</code> , <code>WorkoutSessionEntity</code> , <code>RepScoreEntity</code> , <code>CalibrationProfileDao/Entity</code> , <code>RoomSessionRepository</code> , <code>RoomCalibrationRepository</code> , and the schema JSON. All five Firestore data sources (<code>Workout</code> , <code>Friend</code> , <code>GhostScore</code> , <code>UserProfile</code> , <code>FirestoreWorkoutSession</code>) and <code>SyncingSessionRepository</code> , the Room-first write path. <code>AuthRepository/FirebaseAuthRepository</code> , <code>RepositoryModule</code> . Added the GitHub Actions CI workflow.	46
Xue Li (Claire)	Form scoring and replay. Created and owns <code>FormScorer</code> – the depth/tempo/consistency rule, its calibration gating, the one-sided consistency deviation – with <code>FormScorerTest</code> (12 tests). Created <code>SessionReplayer</code> and <code>SessionReplayerTest</code> , the deterministic re-run recovering each repetition's time window and motion curve from a stored session. Built the Motion Replay presentation layer (<code>MotionReplayUiModels</code> , <code>MotionReplayMapperTest</code>); contributed to <code>SessionDetailScreen</code> , <code>PushupRepDetector</code> , <code>PushupFrameProcessor</code> .	33

Derived from authorship, so pair programming, review, design and device testing do not appear, and commit counts reward granular committers. The middle column is what matters.

5. Rubric justification

Diagrams referenced here are in **Appendix A**, placed after §6 so the figures do not consume the ten pages allowed for this section.

5.1 Implementation — Quality (10)

What we built. One Gradle module whose packages form a strict dependency order, with the domain layer kept free of the Android framework entirely.

Where it lives. See Figure 1. `com.repmate.engine` and `com.repmate.pose` contain **no Android imports** — detectors, the form scorer, calibration derivation, replay and the push-up joint geometry are all plain Kotlin. That is not decoration: it is why 426 of our tests run on the JVM in seconds, with no device, no emulator and no Robolectric.

Separation of concerns is enforced by interfaces, not convention. `SessionRepository`, `FriendRepository`, `LeaderboardRepository` and `CalibrationRepository` are declared in `data/repo/` and implemented in `data/local/`, `data/cloud/` and `data/sync/`. Narrow seams exist specifically so logic is testable without Firebase: `WorkoutUploader` (14 lines), `LeaderboardWriter`, `BestScorePublisher` and `MigrationUploader` each exist so a fake can be substituted. Hilt supplies the wiring (`di/RepositoryModule`, `di/SensorModule`, `di/SafetyModule`, `di/GuestMigrationModule`).

Every screen follows the same shape — one immutable `*UiState`, exposed as a `StateFlow`, with events returned as lambdas. Several are split stateful/stateless (`LeaderboardScreen/LeaderboardContent`, `FriendsScreen/FriendsContent`, `LiveWorkoutContent`) so the stateless half is both previewable and unit-testable.

Error handling is deliberate rather than defensive. The pattern used throughout is an **honest sentinel instead of a plausible lie**: `SessionReplayer.replay()` returns `null` when a session has no stored frames rather than throwing; `FormScorer` sets `rangePercent = NOT_MEASURABLE_RANGE_PERCENT` and the reason "not calibrated: depth not scored" for an uncalibrated user rather than scoring depth against a placeholder; `DeviceSensorSource` completes an empty stream on a device with no accelerometer rather than crashing or fabricating readings. Constructors require blocks reject impossible configurations where they are built — `maxRepDurationMs` must exceed `minRepDurationMs`, `lowThreshold` must sit below `highThreshold`.

Why this meets the criterion. The boundaries are checkable rather than asserted: a reviewer can confirm the domain layer's purity by searching it for `import android.` and finding nothing.

[!GAP] Two namespaces coexist — `com.example.repmate` (holding `MainActivity`, `RepMateApplication`, the Room database and auth) and `com.repmate` (everything else). The `applicationId` is still `com.example.repmate`. This is legacy from the project skeleton. We judged a rename too risky this close to submission for no functional gain, but it is a real inconsistency and we are not hiding it.

5.2 Implementation — Sensors (10)

Which sensors, and why each is necessary.

Sensor	Used for	Why this one
TYPE_ACCELEROMETER	squat and jumping-jack rep detection	carries the whole rep signal; works in a pocket
TYPE_GYROSCOPE	phone-stability gating for camera workouts	rotation is what moves the camera's view
Camera + ML Kit Pose	push-up elbow angle	push-ups barely move the torso — an IMU in a pocket cannot see them
Fused location	map link in the safety escalation SMS	—

Two decisions carry the marks here.

Gravity is deliberately retained. We register `TYPE_ACCELEROMETER`, not `TYPE_LINEAR_ACCELERATION`. Detection runs on the magnitude of acceleration, which is **rotation-invariant** — it reads the same whatever angle the phone sits at in a pocket, a property no single axis has. With gravity included, a still phone parks that magnitude at a known $\sim 9.81 \text{ m/s}^2$, so every threshold is an absolute level around a fixed baseline. Strip gravity out and the resting value becomes ~ 0 , putting every tuned threshold out by 9.81.

The smoothing filter is specified in milliseconds, not samples. This is the most consequential decision in the sensor layer. The filter began life as “25 samples”, which is 250 ms only at 100 Hz. The same 25 samples measured **250 ms** on two $\sim 99 \text{ Hz}$ Pixel recordings, **430 ms** on a $\sim 58 \text{ Hz}$ Samsung recording and **500 ms** at 50 Hz. A moving average’s cutoff frequency is set by its length *in time*, so a sample count silently re-tuned the filter on every device — and with it every threshold downstream, because all of them are measured on the filter’s output. Three separate threshold hunts turned out to be this one bug wearing different hats. The window is now declared as 250 ms and the sample count derived from observed frame timing on every frame, by evicting whatever has aged out of the trailing window.

Lifecycle. `DeviceSensorSource.frames` is a cold `callbackFlow`: collecting it registers the listeners and `awaitClose { stop() }` unregisters them when the collector cancels, including when the `ViewModel` scope dies. A listener left registered keeps the sensor powered for the life of the process, so tying registration to collection is what makes that impossible. Callbacks are delivered to a dedicated `HandlerThread("RepMate-Sensors")`, keeping ~ 100 callbacks a second off the main looper. Timestamps come from `SensorEvent.timestamp` (the monotonic boot clock), never `System.currentTimeMillis()`, which can step backwards mid-workout and give a rep a negative duration.

Noise rejection (Figure 3) is a Schmitt trigger plus four guards, each of which rejects a specific false positive seen in real recordings — a wobble, a rebound, a phone-into-pocket burst, and a state machine stuck open for 36.9 seconds because a parked phone reads 9.81, which sits *between* the two thresholds. Rejections are reported structurally: `RejectedWindow` carries **every** guard a window failed, with measured value against limit, so a missed rep can be traced to the threshold that dropped it and to how far over it was.

Degradation. No gyroscope: the gyro channels stay zero and rep detection continues, because it is accelerometer-driven. No accelerometer: an empty stream, which the screen reports.

[!GAP] **Battery is handled by unregistration alone.** Nothing lowers the sample rate, batches events, or backs off when the screen is off, and no drain measurement was taken.

[!GAP] **StabilityConfig’s thresholds are not measured.** The code says so itself: “*These defaults are a first proposal, not measured values.*” They come from typical sensor-noise figures, not from a recording of a phone propped beside someone doing push-ups, which is the number that matters.

5.3 Implementation — Connectivity (12)

The highest-weighted criterion, and the one with the most to show.

Authentication

Firestore Auth with three paths: Google (via Credential Manager), email/password, and **anonymous guest**. Guests are first-class — a user can complete workouts before ever creating an account — and the guest-to-account path is a credential **link** rather than a new account.

Data model

See Figure 4 for collections and their shapes. Seven collections, each with its own rule block in `firestore.rules` (230 lines).

Security rules — the part worth reading

The rules do real work rather than restating `request.auth != null`:

- **Unique display names without a uniqueness query.** `usernames/{name}` is a *claim ticket* whose document id is the name lowercased. A second claim is a create on an existing document, and **there is no update rule at all**, so it is refused. That is the uniqueness guarantee — no query, no race.
- **Cross-document invariants.** The claim rule uses `getAfter()` to assert that the user's profile `displayName` *after the write* lowercases to the claimed id, forcing the claim and the profile change into one batch. The rename rule additionally refuses to leave a user holding their old claim, so a rename cannot hoard names.
- **Field allow-lists.** Every writable document is pinned with `.keys().hasOnly([...]) and .hasAll([...])`, so neither an extra field nor a missing one is accepted. `leaderboard` also requires `points is int, 0 <= points <= 100000` and `updatedAt == request.time`, so a modified client cannot write itself to the top or forge a timestamp.
- **Enumeration is denied.** `usernames` and `ghostScores` are `allow get but allow list: if false` — readable one document at a time, never enumerable. That is why the Friends leaderboard reads `ghostScores` per friend, and why the Global leaderboard needed its own collection.
- **Guests are excluded by provider**, not by naming convention: `isRealAccount()` checks `request.auth.token.firebase.s...` `!= 'anonymous'`.
- **Name validation is ASCII-only on purpose.** If `lower()` were Unicode-aware, a look-alike such as the Kelvin sign (U+212A, which lowercases to “k”) could map onto an id someone else owns.

These rules are **tested**: 68 cases in `tests/firestore.rules.test.cjs`, run against the Firestore emulator, and **run in CI on every pull request** — not merely deployed and hoped over.

Offline-first behaviour

Room is the source of truth. `SyncingSessionRepository.save()` writes Room first and unconditionally, then attempts Firestore; a cloud failure is logged and the workout still completes. Nothing a user does is lost to a dropped connection.

Reconciliation is by recompute, not a retry queue (Figure 6). `LeaderboardSync` and `LocalBestScoreSync` recompute the best session per exercise *from Room* and republish. This is idempotent — a workout finished offline is included next time either runs, and running twice writes the same number — so it needs no durable outbox, no retry bookkeeping and no dead-letter handling. Both run from two places: after a workout saves, and after History restores from the cloud.

Conflict handling. Ghost scores use a **Firestore transaction**, so a lower score cannot overwrite a higher one when two devices race. Everything else is last-write-wins over idempotent recomputation, which is safe precisely because the written value is a pure function of local history rather than a delta.

Interrupted guest migration resumes. Merging a guest's history into a real account can be cut short by a crash, a kill or a lost connection. `PendingMigrationStore` keeps a set of pending migrations keyed by guest UID in `DataStore`, and `GuestMigrationLauncher.resumeOnAppStart()` walks the queue on every launch, a no-op when it is empty.

Real-time listeners. Friend requests, the friends list and both leaderboard tabs are Firestore snapshot listeners, so a request accepted on one device appears on the other without a refresh.

[!GAP] **Firestore offline persistence is never explicitly configured.** It works because the Android SDK enables the disk cache by default. We rely on a default we never stated.

[!GAP] **The app never tells the user it is offline.** There is no `ConnectivityManager` Or `NetworkCallback` anywhere in the codebase. Work is preserved correctly, but silently — a user with no connection sees no banner and no pending indicator.

5.4 Implementation — Responsiveness (6)

What is off the main thread. Sensor callbacks are delivered to a dedicated `HandlerThread`. Camera frames are analysed on `Executors.newSingleThreadExecutor()` with CameraX's `STRATEGY_KEEP_ONLY_LATEST` — real back-pressure, where a frame arriving while the previous one is still being analysed is dropped rather than

queued, so the pose pipeline cannot build a backlog. Post-workout Firestore writes are launched in an injected `@ApplicationScope` (`SupervisorJob + Dispatchers.IO`) explicitly so the finish screen does not wait on up to three network round trips. Room and DataStore suspend functions dispatch internally.

Back-pressure on the sensor stream is handled at the source rather than with an operator. The tick loop keeps a running deadline instead of `delay(20)` in a loop, which accumulates drift, and when it falls behind it **resyncs to now rather than sprinting to catch up**, because a burst of back-to-back frames would be a worse lie than a missing one. A frame is emitted only when the accelerometer timestamp has actually advanced, so a stream running slower than the tick never re-emits a stale reading as new motion.

A long set cannot exhaust memory: replay capture stops at `MAX_REPLAY_FRAMES = 30,000` (ten minutes at the sample rate) rather than growing without bound, and stops capturing rather than dropping the oldest frames, so what is kept is a true prefix of the set.

[!GAP] **The IMU detector runs on the main thread.** `LiveWorkoutViewModel` collects `sensorSource.frames` in `viewModelScope`, which is `Dispatchers.Main.immediate`, and because `callbackFlow`'s producer inherits the collector's context, both the 50 Hz tick loop and every `detector.process()` call run there. Per-frame cost is $O(1)$ — a circular-buffer push, an eviction and a handful of comparisons — and we have observed no dropped frames or jank on a Pixel 10a. But there is no `flowOn`, no buffer and no `conflate` on that path, so this is a property we have *observed*, not one the code *guarantees*.

[!GAP] **No latency or frame-rate figure has been measured.** Nothing in the code, the tests or the documentation quotes a number for per-frame processing cost or end-to-end rep-to-pixel latency. The claims above are structural, not empirical.

5.5 Implementation — Technical depth (6)

Three detectors, three different signal problems.

Squats (Figure 3) run on smoothed accelerometer magnitude through a Schmitt trigger. The hysteresis gap between 10.15 and 9.7 is what stops a signal hovering at one value from rattling the state machine. Every guard figure is derived from measured data rather than chosen for roundness: `minRepDurationMs` of 550 ms sits 65 ms below the shortest real rep in the corpus (615 ms); `maxRepDurationMs` of 3000 ms is near twice the longest (1573 ms); `cooldownMs` of 500 ms leaves headroom above the tightest genuine gap (917 ms).

Jumping jacks are not one burst but **two impacts per rep** — the jump out and the landing — so `JumpingJackRepDetector` pairs bursts rather than counting them, at a paced `TARGET_BPM = 52`. The metro-nome reads that same constant, so audio, the visual pulse and the detector cannot drift apart: they are one number, not three copies of it.

Push-ups run on elbow angle from ML Kit pose, filtered with a **median, not a mean**. A single mislabelled landmark is removed outright by a median; a mean only attenuates it, and one frame at 40° would still drag 168° down to 104° and count a phantom rep.

Calibration is derived, not guessed. Across four recorded participants, squat amplitude spans **7.1×** between the softest and the strongest while both squat correctly, so a single global threshold is set by whoever squats softest. `CalibrationProfile` derives four per-user thresholds from a five-rep set, and each margin is justified against measured drift between the first and last five reps of recorded sets — `AMPLITUDE_MARGIN` 0.65 against a worst observed softening of 0.771; `MAX_DURATION_MARGIN` 2.2 as the geometric centre of a band bounded below by 1.214 and above by 4.03.

Replay determinism. `SessionReplayer` re-runs a session's stored raw frames through the same detector and scorer used live. It is deterministic because the moving average is **causal** — it never looks at future samples, so it behaves identically live and in replay — the detectors are pure and read no clock, and the same `CalibrationProfile` is passed back in. What would break it, stated explicitly: changing `smoothingWindowMs` or any threshold; frames arriving out of chronological order; a non-monotonic timestamp source; or replaying under a different profile.

Honest sentinels over plausible numbers. `RepScore.pauseSeconds` is always 0f and documented as a placeholder. Because magnitude is direction-blind, the detector cannot distinguish descent from ascent, so no

timestamp exists for “reached the bottom” and there is nothing to measure a dwell against. We report zero and say why rather than inventing a figure.

[!GAP] **The strongest thresholds rest on thin evidence, and the code says so.** `minAmplitude` separates wobbles from reps unaided, on a band 0.15 m/s^2 wide, **both edges of which come from a single participant**. The jumping-jack defaults come from two recordings of one person on one phone. The push-up thresholds come from one person, one session, one camera placement. More participants would either widen these bands into something defensible or show that no fixed threshold generalises — and that second answer argues for per-user calibration everywhere, which is why squats and jumping jacks already have it.

5.6 UI — Appeal (4)

A single deliberate visual identity: charcoal #121212 with an acid-lime #C6F135 accent, applied through one Material 3 ColorScheme and one Typography. Three shared components — `RepMateCard`, `RepMateButton` (variants `Solid` and `Ghost`) and `RepMateTopBar` — give every screen the same card radius, border and header treatment.

Charcoal rather than pure black is considered, not accidental: pure black crushes contrast on OLED and reads harshly next to a bright accent. Dynamic (wallpaper-derived) colour is **deliberately not enabled**, because it would replace the lime accent with a device-specific colour and dissolve the identity the palette exists to create.

`RepMateTopBar` was extracted from two private, identical copies in `GhostDuelScreen` and `SessionDetailScreen`, so a fourth screen cannot quietly drift from the other three.

5.7 UI — Guidelines (6)

Material Design 3. `MaterialTheme` with `darkColorScheme/lightColorScheme`, custom Typography and Shapes; stock M3 `NavigationBar`, `Switch`, `SnackBar`, `AlertDialog`, `Icon` and `IconButton`. The one screen that had drifted onto Material defaults — Friends, which used `Scaffold/TopAppBar` and referenced colour roles our theme does not define, so it rendered in baseline Material purple — was rebuilt on the shared components in commit 00cb8f3.

Android accessibility guidelines, specifically:

- *Provide content labels.* 36 `contentDescriptions`, and — equally important — **19 explicitly set to null** on decorative icons, which is what the guideline actually asks for: a screen reader should not announce an ornament.
- *Announce the right role.* Interactive elements declare semantics rather than relying on a bare `clickable: Role.Button, Role.Tab` (leaderboard tabs, so `TalkBack` says which is selected), `Role.Switch` and `Role.Checkbox`, with `onClickLabel` on clickables.
- *Support text scaling.* All 25 type sizes are declared in `sp`, with **no font size anywhere declared in dp**, so system font scaling works.
- *Do not rely on colour alone.* The current user’s leaderboard row carries the text “you” as well as a tint; a history session that recorded no reps shows “—” rather than a colour change.
- *Respect the user’s theme.* Dark mode is an in-app preference backed by `DataStore`, with `LocalRepMateDarkTheme` so resource-picking composables stay consistent with it.

[!GAP] **Touch targets are large by layout, not by constraint.** Only five places in the UI set an explicit minimum size and nothing calls `minimumInteractiveComponentSize()`. We believe all targets meet 48 dp but have not verified it.

[!GAP] **No contrast ratios have been computed and no TalkBack pass has been performed.** Roles and labels are declared; nobody has navigated a screen with a screen reader to confirm the result is usable. Declaring semantics is necessary but not sufficient, and we should not claim an accessibility result we have not observed.

[!GAP] **The light theme has never been run on a device** — it exists and is exercised by Compose previews only.

5.8 UI — Flow (6)

Eighteen destinations (Figure 5). Three decisions are worth defending:

Launch routing lives in exactly one function. `RepMateDestinations.afterAuth()` decides where a signed-in user lands — display-name gate, onboarding or Home — and is called both after an interactive sign-in and at launch for a restored session. One rule, one implementation, so the two paths cannot disagree. The system splash is held until `StartupViewModel` has resolved the destination, so a returning user never sees Welcome flash before Home.

Calibration sits on the path into a workout, not in a settings screen. Choosing squats on Home routes through `calibration/{exerciseType}/{entryPoint}`, carrying an entry point so the same screen knows whether it was reached as a first-time gate or as a deliberate recalibration from Profile.

One screen, two modes, via an optional query parameter. `session_detail/{id}?postWorkout=` is the post-workout summary when reached from a finished workout and plain detail when reached from History. The parameter is omitted when false, so History's route is byte-identical to what it always was.

5.9 UI — Language (4)

Feedback is phrased as observation, not verdict. A rep reads “good depth, slower than usual” or “not deep enough, good tempo” — specific and actionable, assembled from the individual reasons `FormScorer` emits rather than reduced to a bare score.

The clearest language decision is **refusing to say something we cannot know**. An uncalibrated user is told “not calibrated: depth not scored” instead of being shown a depth score computed against a placeholder. Given measured amplitude spans about 9× between real users, a fixed reference would tell a strong squatter they are permanently “too deep” and a light squatter they are permanently “not deep enough” — and someone might change an already-correct squat to chase a misleading number. Likewise a session that recorded no reps shows “–”, not “0.0”, because 0.0 reads as a real and catastrophic score.

Section labels are lowercase and quiet (“your friends”, “ghost duel”, “score trend”); numbers carry their units (“197 pts”, “8 reps”).

5.10 UI — Reactiveness (6)

Every screen renders from one immutable `UiState` exposed as a `StateFlow` and collected with `collectAsStateWithLifecycle`, so recomposition is scoped and collection stops when the screen is not resumed. There is no mutable shared UI state and no manual invalidation.

During a live workout the rep counter, the elapsed timer and the per-rep feedback all update from the same emission, so the number on screen, the haptic buzz and the spoken count cannot disagree. Firestore-backed screens are driven by snapshot listeners rather than polling, so a friend request accepted elsewhere appears without a refresh. Settings toggles write to `DataStore` and are read back as flows, so the theme switch takes effect immediately across the app rather than at next launch.

Loading and empty states are modelled explicitly in the state classes rather than inferred from a null, so a screen with no data says so instead of rendering blank.

5.11 Innovation — Novelty (3)

The ghost duel. Rather than a leaderboard number, a user races a friend's *best recorded set* — same exercise, their score to beat, shown live. It reuses the scoring pipeline rather than adding a parallel one.

Motion replay. After a set, the user sees the actual smoothed accelerometer curve for each rep with their calibration band drawn behind it. Most fitness apps show a count; this shows the signal the count came from.

A metronome as part of the detector's contract. Jumping-jack counting *requires* a paced cadence, so rather than hiding that assumption the app supplies the pace, and the detector and the audio share one constant.

5.12 Innovation — Surprise (3)

The intended moment of surprise is Motion Replay: a user expecting a rep counter instead sees the waveform their body produced, with their own calibrated range behind it, and can see *why* rep 7 scored lower than rep

6.

The second is the safety check-in — a fitness app that notices you have not confirmed you are all right after a session, and escalates to a contact with your last known location.

[!GAP] This criterion is argued rather than evidenced; no artefact in the repository demonstrates “surprise”. It depends entirely on how the demo video stages these two moments.

5.13 Innovation — Tech Knowledge (4)

The techniques below are used because the problem demanded them, and each is documented at its definition with the measurement that justified it:

- **Schmitt trigger / hysteresis** in all three detectors, so a signal hovering at a threshold is not counted repeatedly.
- **Rotation-invariant magnitude**, making detection independent of phone orientation.
- **Causal, time-specified moving average**, so the cutoff frequency is a property of the filter rather than of the device’s sample rate — and so replay is identical to live.
- **Median filtering** on the pose angle, chosen over a mean because it removes outliers outright rather than attenuating them.
- **O(1) circular-buffer filtering** with primitive arrays and a running sum, so the hot path allocates nothing per frame.
- **Transactional max-write** for ghost scores, and **idempotent recompute** instead of a retry queue for sync.
- **Claim-ticket uniqueness** in security rules, using document-id collision rather than a query.

5.14 Innovation — Cross-Disciplinary (3)

The genuine cross-disciplinary content is **digital signal processing applied to a biomechanical signal**: time-domain filter design, hysteresis, outlier-robust statistics and the rotation-invariance argument are signal-processing results rather than software-engineering ones, and they are what makes the counting work.

There is also a **human-factors** thread with a real engineering consequence: the finding that squat amplitude spans roughly 9× between individuals who are all squatting correctly is why the app calibrates per user and refuses to score depth without a profile. The design accommodates human variation instead of asserting a norm.

[!GAP] **There is no exercise-science or biomechanics literature behind the form model.** We searched the codebase to confirm this before writing. The depth, tempo and consistency weights (4, 3 and 1.5) are a first pass, documented in FormScorer as “a reasonable starting point, not a value agreed with the rest of the team”. Scoring is *self-relative* — your reps against your own calibration — which is defensible engineering but is not exercise science. Claiming biomechanical grounding would be inventing it. Closing this properly means citing literature for a depth or tempo standard and re-deriving the weights from it.

5.15 Innovation — Impact (3)

The app addresses unsupervised exercise, where the failure mode is not laziness but doing the movement badly without knowing. Per-rep feedback phrased as observation, calibrated to the individual, is aimed at exactly that.

Two choices widen who can use it. `minSdk 26` (Android 8.0, 2017) keeps the app on old hardware rather than only on recent flagships. **Guest mode** lets someone complete a workout and see their scores without creating an account, and the guest-to-account path preserves their history if they later sign up.

The **safety check-in** targets a specific risk: exercising alone. The app sends a notification after a session and, if it is not acknowledged, escalates to a nominated contact by SMS with the last known location. The acknowledgement flag is re-checked at send time, so an alert is not sent to someone who confirmed they were fine while the timer was pending.

6. AI tool acknowledgement

Generative AI tools were used during this project. In accordance with the assignment brief, their use is disclosed below. A running log is also maintained in `AI_USAGE_LOG.md` in the repository root. Every entry records what was generated, by whom and how the output was reviewed before it entered the codebase; no output was committed without a named member reviewing it.

Tool	Where used	Example of use	How the output was reviewed
[FILL]	[FILL]	[FILL]	[FILL]
[FILL]	[FILL]	[FILL]	[FILL]
[FILL]	[FILL]	[FILL]	[FILL]
[FILL]	[FILL]	[FILL]	[FILL]
[FILL]	[FILL]	[FILL]	[FILL]

[!GAP] This table is deliberately empty. Each member must complete their own rows — nobody else can accurately describe how a third party reviewed an AI output. Note that 30 of the repository's commits carry Co-Authored-By trailers naming an AI assistant, all dated on or before 2026-09-20; `git log --all --grep="Co-Authored-By"` lists them and is a useful starting point.

Appendix A — Diagrams

Placed here so the figures do not consume the page allowance for \$5. Each is referenced from the section it supports. Mermaid sources are in report/diagrams/*.md.

Figure 1 — System architecture (\$5.1)

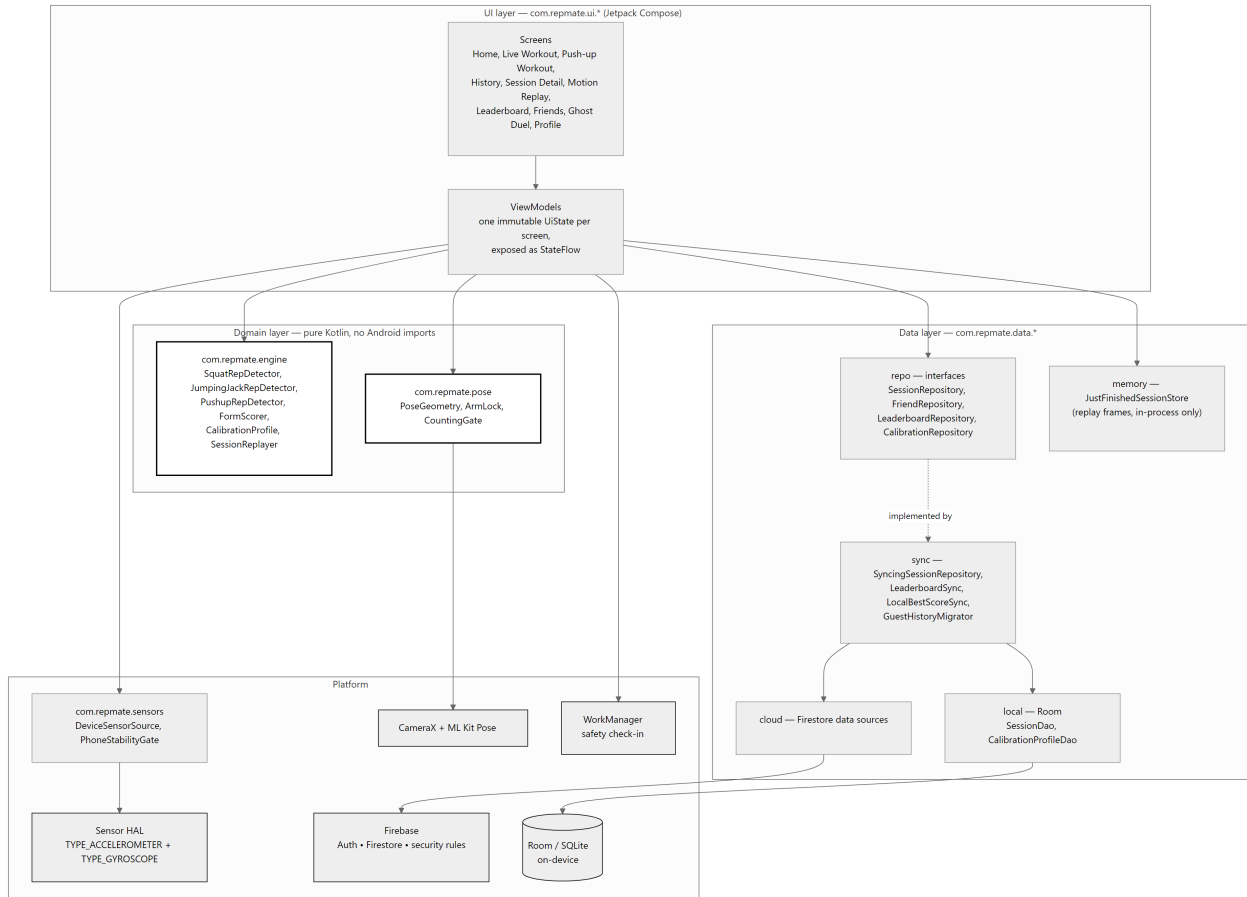


Figure 1: System architecture

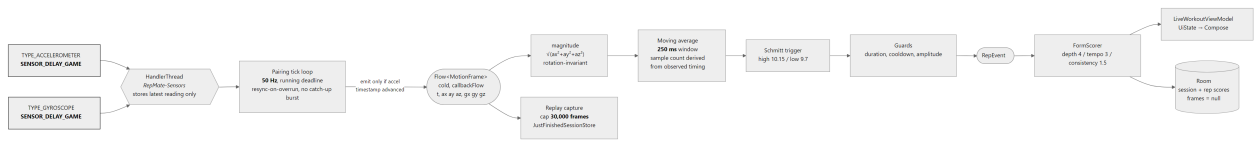
Figure 2 — Sensor data pipeline (§5.2)

Figure 2: Sensor data pipeline

Figure 3 — Squat rep-counting state machine (§5.2, §5.5)

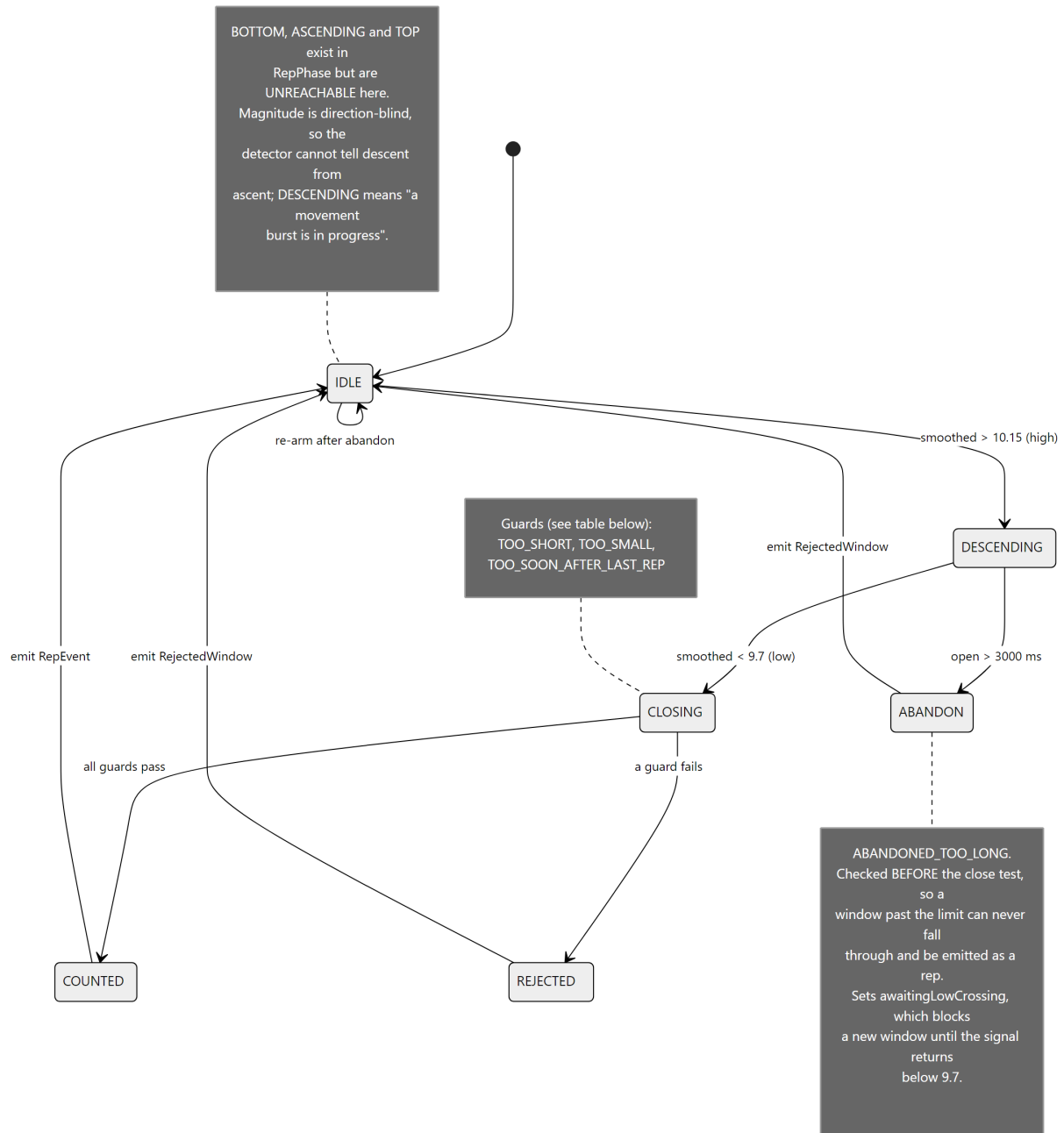


Figure 3: Rep-counting state machine

Figure 4 — Cloud data model and sync (§5.3)

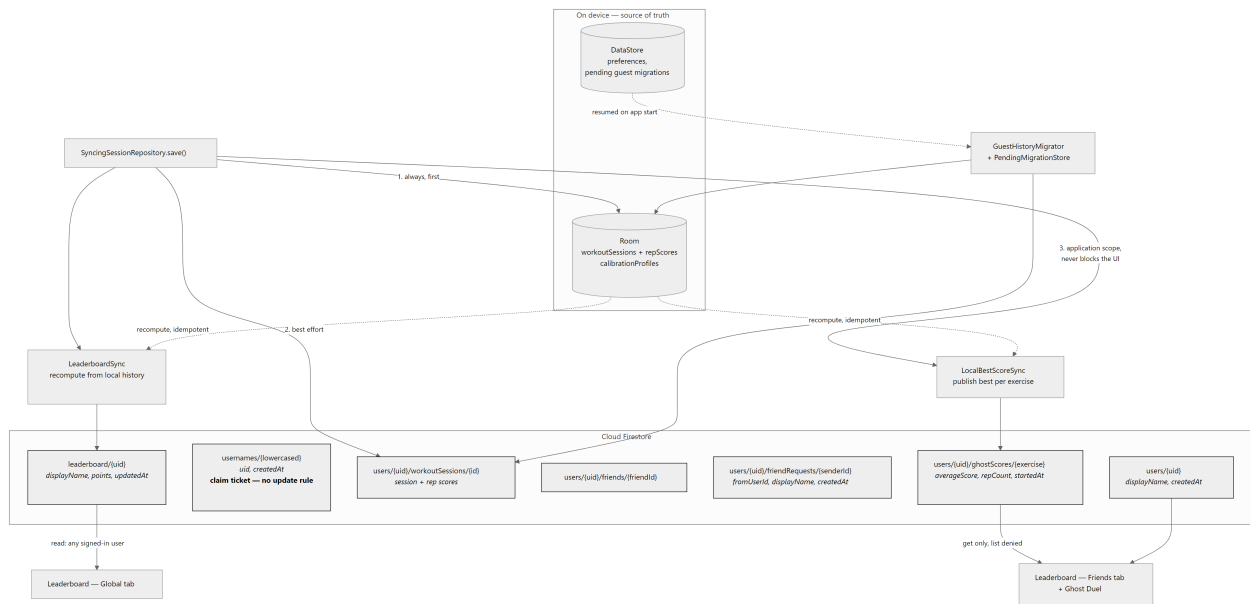


Figure 4: Cloud data model and sync

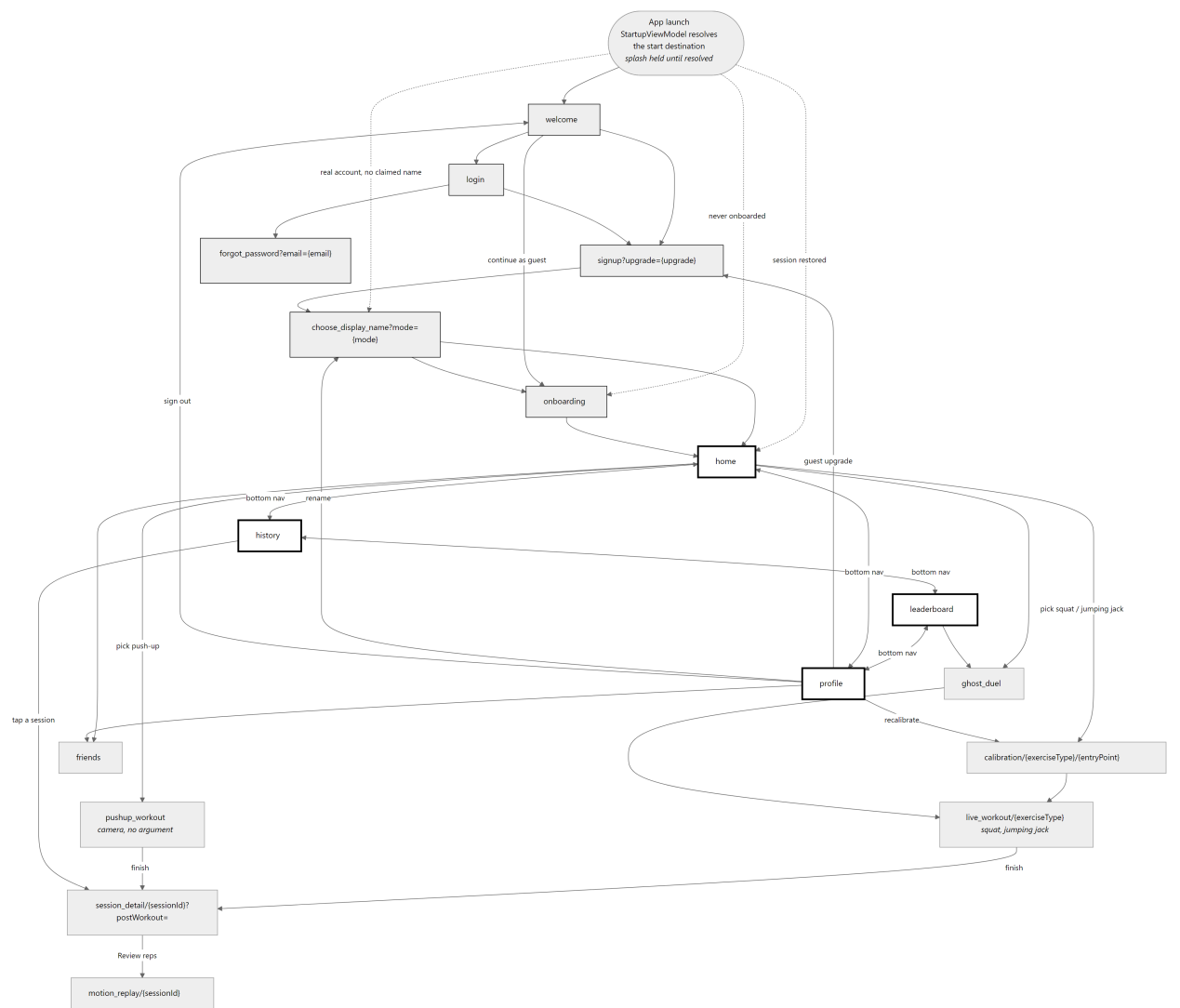


Figure 6 — End-to-end sequence: workout to leaderboard (§5.3)

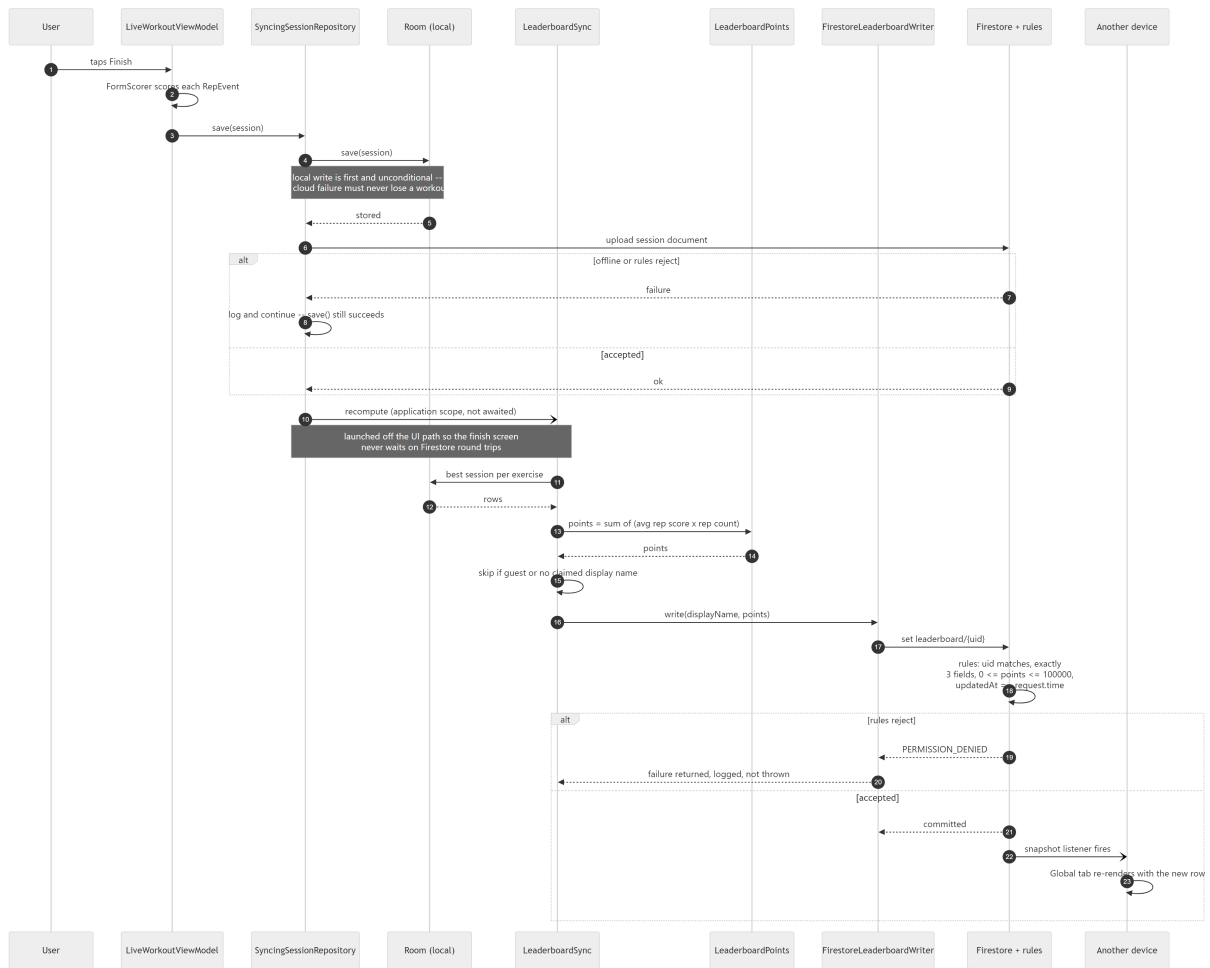


Figure 6: Leaderboard write sequence

Appendix B — Test and verification summary

Suite	Count	Status	How to reproduce
JVM unit tests	426	0 failures, 0 errors, 0 skipped, 0 @Ignore	<code>./gradlew testDebugUnitTest</code>
Instrumented tests	47 across 8 files	compile in CI; 30 verified running on a Pixel 10a	<code>./gradlew connectedDebugAndroidTest</code>
Firestore rules tests	68 cases	pass against the emulator, run in CI	<code>npx firebase emulators:exec --only firestore "node --test tests/firestore.rules.test.cjs"</code>

Engine tests are backed by real recordings, not synthetic signals. Six recorded traces ship as test resources in `app/src/test/resources/traces/`, each paired with a `.expect` file holding ground truth — how many reps the recording contains, and how many events the whole file should produce including deliberate non-rep bursts. `TraceLibrary` discovers any new `.csv` + `.expect` pair automatically. Raw unstripped exports are archived in `traces/raw/` as zips.

Continuous integration (`.github/workflows/android-ci.yml`) runs on every pull request and every push to main: `assembleDebug`, then `testDebugUnitTest`, then `compileDebugAndroidTestKotlin`, then the Firestore rules tests against the emulator on JDK 21.

The instrumented-test compile step was added after discovering that `androidTest` had silently stopped compiling — a production method was renamed and the test file kept calling the old name. A source set that does not compile takes **every** test in it down, so the Room, DataStore and safety-worker tests were unrunnable too, and neither `assembleDebug` nor `testDebugUnitTest` noticed. The gate was verified by reintroducing the breakage and confirming the step fails.

[!GAP] **8 of the 47 instrumented tests have never been executed.** `FirestoreFriendRepositoryTest` was rewritten for the friend-request flow and compiles, but running it needs the Firestore and Auth emulators plus disposable account credentials, which we have not had set up at the same time as a connected device. They are not claimed as passing.

[!GAP] **CI compiles the instrumented tests but does not run them.** Doing so needs an AVD in CI plus the emulators and credentials.

Appendix C — Known limitations

Collected from the [!GAP] markers above so a marker can see them in one place. We judged it better to state these than to write around them.

#	Limitation	Section
1	Two package namespaces coexist; <code>applicationId</code> is still <code>com.example.repmate</code>	§5.1
2	Battery handling is unregistration only; no drain measured	§5.2
3	<code>StabilityConfig</code> thresholds are proposed, not measured	§5.2
4	Firestore offline persistence relies on an unstated SDK default	§5.3
5	The app never indicates to the user that it is offline	§5.3

#	Limitation	Section
6	The IMU detector runs on the main thread with no <code>flowOn</code>	§5.4
7	No latency or frame-rate figure has been measured	§5.4
8	Key thresholds rest on one or two participants	§5.5
9	<code>RepScore.pauseSeconds</code> is always 0 — not derivable from magnitude	§5.5
10	Touch targets unverified; no contrast ratios; no <code>TalkBack</code> pass	§5.7
11	Light theme never run on a device	§5.7
12	No exercise-science grounding for the form-scoring weights	§5.14
13	8 instrumented tests have never been executed	App. B